

ASSIGNMENT 1: Linear Algebra - MCQ Assessment

Section A: MCQs

Question 1

A system is consistent if:

- a) $\det(A) = 0$ b) $\text{rank}(A) = \text{rank}([A|B])$ c) $\text{rank}(A) \neq \text{rank}([A|B])$ d) A is identity

Answer: b) $\text{rank}(A) = \text{rank}([A|B])$

A system is consistent exactly when the coefficient and augmented matrices have equal rank.

Question 2

If $\text{rank}(A) = \text{rank}([A|B]) = \text{number of variables}$, the system has:

- a) Infinite solutions b) No solution c) Unique solution d) Two solutions

Answer: c) Unique solution

Equal ranks matching the number of variables give a consistent system with a unique solution.

Question 3

A homogeneous system always has:

- a) No solution b) At least one solution c) Exactly one solution d) Infinite solutions

Answer: b) At least one solution

The zero vector is always a solution of a homogeneous system.

Question 4

If a homogeneous system has $\text{rank} < \text{number of variables}$, then:

- a) Unique solution b) Infinite solutions c) No solution d) Cannot determine

Answer: b) Infinite solutions

A homogeneous system is consistent; rank below the number of variables leaves free variables.

Question 5

Which condition implies inconsistency?

- a) $\text{rank}(A) = \text{rank}([A|B])$ b) $\text{rank}(A) \neq \text{rank}([A|B])$ c) $\det(A) \neq 0$ d) A is diagonal

Answer: b) $\text{rank}(A) \neq \text{rank}([A|B])$

Unequal coefficient and augmented ranks are the consistency criterion for no solution.

Question 6

If $\det(A) \neq 0$, then:

- a) A is singular b) A has no inverse c) A is invertible d) A has infinite solutions

Answer: c) A is invertible

A nonzero determinant means the square matrix is nonsingular and invertible.

Question 7

Eigenvalues are roots of:

- a) $AX = B$ b) $\det(A - \lambda I) = 0$ c) $\text{rank}(A) = 0$ d) $A^2 = I$

Answer: b) $\det(A - \lambda I) = 0$

Eigenvalues satisfy the characteristic equation $\det(A - \lambda I) = 0$.

Question 8

If λ is an eigenvalue, then:

- a) $AX = \lambda X$ b) $AX = 0$ c) $AX = B$ d) $A = I$

Answer: a) $AX = \lambda X$

For an eigenvector X , $AX = \lambda X$, with X nonzero.

Question 9

For identity matrix, eigenvalues are:

- a) 0 b) 1 c) -1 d) Cannot determine

Answer: b) 1

The identity matrix maps every vector to itself, so every eigenvalue is 1.

Question 10

Trace of a matrix equals:

- a) Product of eigenvalues b) Sum of diagonal elements c) Rank d) Determinant

Answer: b) Sum of diagonal elements

The trace is the sum of the main-diagonal entries.

Question 11

Rank of a matrix is:

- a) Number of rows b) Number of columns c) Maximum number of linearly independent rows/columns d) Determinant

Answer: c) Maximum number of linearly independent rows/columns

Rank is the maximum number of linearly independent rows or columns.

Question 12

If two rows are identical, rank will:

- a) Increase b) Decrease c) Remain same d) Become zero

Answer: b) Decrease

Identical rows are linearly dependent, so they do not provide independent row directions.

Question 13

A matrix with all zeros has rank:

- a) 1 b) 0 c) n d) Cannot determine

Answer: b) 0

The zero matrix has no nonzero independent rows or columns.

Question 14

If rows are linearly dependent:

- a) Rank is full b) Rank is reduced c) Rank is maximum d) Determinant is 1

Answer: b) Rank is reduced

Dependent rows reduce the number of independent rows and therefore the rank.

Question 15

Maximum rank of an $m \times n$ matrix is:

- a) $m + n$ b) $m \times n$ c) $\min(m, n)$ d) $\max(m, n)$

Answer: c) $\min(m, n)$

Rank cannot exceed either the number of rows or columns.

Question 16

If rank < number of variables, system has:

- a) Unique solution b) Infinite solutions c) No solution d) Cannot say

Answer: d) Cannot say

Rank(A) alone is insufficient: consistency also depends on $\text{rank}([A|B])$. If consistent, $\text{rank}(A) < n$ gives infinitely many solutions; otherwise there is no solution.

Question 17

Which operation does NOT change rank?

- a) Row swap b) Row scaling c) Row addition d) Changing constants

Answer: d) Changing constants

Elementary row operations (swap, nonzero scaling, and row addition) preserve rank. Changing constants is not a row operation on A.

Question 18

Eigenvectors corresponding to distinct eigenvalues are:

- a) Dependent b) Independent c) Zero d) Equal

Answer: b) Independent

Eigenvectors belonging to distinct eigenvalues are linearly independent.

Question 19

Determinant of triangular matrix is:

- a) Sum of diagonal b) Product of diagonal c) Zero d) One

Answer: b) Product of diagonal

The determinant of a triangular matrix is the product of its diagonal entries.

Question 20

If determinant is zero, system:

- a) Always inconsistent b) Always unique c) May be consistent or inconsistent d) Always infinite

Answer: c) May be consistent or inconsistent

A singular coefficient matrix can yield no solution or infinitely many, depending on B.

Question 21

A square matrix is invertible if:

- a) $\text{rank} < n$ b) $\det = 0$ c) $\det \neq 0$ d) $\text{rank} = 0$

Answer: c) $\det \neq 0$

A square matrix is invertible exactly when its determinant is nonzero.

Question 22

If $AX = 0$ has only trivial solution:

- a) $\det(A)=0$ b) $\det(A)\neq 0$ c) $\text{rank} < n$ d) A is singular

Answer: b) $\det(A)\neq 0$

For a square matrix, only the trivial nullspace means full rank and a nonzero determinant.

Question 23

Characteristic equation is:

- a) $AX = B$ b) $\det(A - \lambda I) = 0$ c) $\text{rank}(A) = n$ d) $A^2 = 0$

Answer: b) $\det(A - \lambda I) = 0$

The characteristic equation is $\det(A - \lambda I) = 0$.

Question 24

If $\lambda = 0$ is eigenvalue:

- a) Matrix invertible b) Matrix singular c) Matrix identity d) Rank full

Answer: b) Matrix singular

If zero is an eigenvalue, the matrix has a nontrivial nullspace and is singular.

Question 25

A system with parallel planes:

- a) Unique solution b) Infinite solutions c) No solution d) One solution

Answer: c) No solution

Distinct parallel planes do not intersect, so the system has no common solution.

Question 26

If three planes intersect at a point:

- a) No solution b) Infinite solutions c) Unique solution d) Two solutions

Answer: c) Unique solution

A single point of intersection is exactly one solution.

Question 27

If all planes coincide:

- a) No solution b) Infinite solutions c) Unique solution d) Two solutions

Answer: b) Infinite solutions

Coincident planes share every point on that plane, giving infinitely many solutions.

Question 28

Rank of identity matrix ($n \times n$):

- a) 1 b) n c) 0 d) $n-1$

Answer: b) n

An n by n identity matrix has n independent rows and columns.

Question 29

If one row is multiple of another:

- a) Rank increases b) Rank decreases c) Rank unchanged d) Rank infinite

Answer: b) Rank decreases

A row that is a multiple of another is dependent, reducing the maximum independent-row count.

Question 30

If $AX = B$ is inconsistent:

- a) $\text{rank}(A) = \text{rank}([A|B])$ b) $\text{rank}(A) < \text{rank}([A|B])$ c) $\text{rank}(A) > \text{rank}([A|B])$ d) $\text{rank} = 0$

Answer: b) $\text{rank}(A) < \text{rank}([A|B])$

Inconsistency is characterized by $\text{rank}(A) < \text{rank}([A|B])$.

Question 31

Eigenvalues of diagonal matrix are:

- a) Zero b) Diagonal elements c) Sum d) Product

Answer: b) Diagonal elements

For a diagonal matrix, the diagonal entries are its eigenvalues.

Question 32

If A is symmetric:

- a) Eigenvalues complex b) Eigenvalues real c) No eigenvalues d) Rank zero

Answer: b) Eigenvalues real

Every real symmetric matrix has real eigenvalues.

Question 33

If $A^2 = A$, eigenvalues are:

- a) 0 or 1 b) Only 1 c) Only 0 d) Any number

Answer: a) 0 or 1

If $Ax = \lambda x$ and $A^2 = A$, then $\lambda^2 = \lambda$, so λ is 0 or 1.

Question 34

Rank equals number of:

- a) Zero rows b) Non-zero independent rows c) Columns d) Determinant

Answer: b) Non-zero independent rows

Rank is the number of nonzero independent rows in row-echelon form.

Question 35

If system has infinite solutions:

- a) rank = variables b) rank < variables c) rank > variables d) rank = 0

Answer: b) rank < variables

Infinitely many solutions require a free variable, so rank(A) is below the number of variables.

Question 36

If two eigenvalues are equal:

- a) Eigenvectors always same b) Eigenvectors may differ c) No eigenvectors d) Matrix zero

Answer: b) Eigenvectors may differ

Repeated eigenvalues can have an eigenspace of dimension less than their multiplicity.

Question 37

A zero determinant implies:

- a) Unique solution b) Infinite or no solution c) Always no solution d) Always unique

Answer: b) Infinite or no solution

A zero determinant makes the coefficient matrix singular; depending on the right-hand side, there may be infinitely many or no solutions.

Question 38

If matrix is orthogonal:

- a) $A^{-1} = A^T$ b) $A = 0$ c) $\det = 0$ d) rank = 0

Answer: a) $A^{-1} = A^T$

An orthogonal matrix satisfies $A^T A = I$, hence $A^{-1} = A^T$.

Question 39

Trace equals:

- a) Sum of eigenvalues b) Product of eigenvalues c) Rank d) Determinant

Answer: a) Sum of eigenvalues

The trace equals the sum of eigenvalues, counting algebraic multiplicity.

Question 40

If eigenvalue lambda repeated n times:

- a) Always n independent eigenvectors b) May have fewer eigenvectors c) No eigenvectors d) Rank zero

Answer: b) May have fewer eigenvectors

A repeated eigenvalue need not have a full set of independent eigenvectors.

Question 41

If matrix rank is 1:

- a) All rows independent b) Only one independent row c) Zero matrix d) Full rank

Answer: b) Only one independent row

Rank one means there is exactly one independent row direction.

Question 42

If $\det(A)=0$, inverse:

- a) Exists b) Does not exist c) Identity d) Diagonal

Answer: b) Does not exist

A square matrix with zero determinant is singular and has no inverse.

Question 43

If $AX=B$ has no solution:

- a) Consistent b) Inconsistent c) Unique d) Infinite

Answer: b) Inconsistent

A system with no solution is, by definition, inconsistent.

Question 44

Eigenvector cannot be:

- a) Zero vector b) Non-zero vector c) Unit vector d) Column vector

Answer: a) Zero vector

An eigenvector must be nonzero; the zero vector cannot define an eigenvector.

Question 45

If matrix is scalar matrix:

- a) All eigenvalues same b) All eigenvalues zero c) Rank zero d) No eigenvalues

Answer: a) All eigenvalues same

A scalar matrix cI has the same eigenvalue c in every direction.

Question 46

If A is upper triangular, eigenvalues are:

- a) First row b) Diagonal elements c) Sum of rows d) Product of rows

Answer: b) Diagonal elements

The eigenvalues of a triangular matrix are its diagonal entries.

Question 47

If rank = number of variables:

- a) Infinite solutions b) Unique solution c) No solution d) Cannot determine

Answer: d) Cannot determine

$\text{Rank}(A) = n$ guarantees uniqueness only if the system is consistent. Without $\text{rank}([A|B])$, consistency cannot be determined.

Question 48

If $AX=0$ has infinite solutions:

- a) $\det \neq 0$ b) $\det = 0$ c) $\text{rank} = n$ d) unique

Answer: b) $\det = 0$

Infinitely many solutions to $A X = 0$ mean a nontrivial nullspace, so A is singular and $\det(A) = 0$.

Question 49

If system has more equations than variables:

- a) Always inconsistent b) Always consistent c) May or may not be consistent d) Always infinite

Answer: c) May or may not be consistent

Having more equations than variables does not decide consistency; the equations may or may not agree.

Question 50

If vectors are linearly independent:

- a) One is multiple of another b) No vector is combination of others c) All are zero d) Rank zero

Answer: b) No vector is combination of others

Linear independence means no vector in the set is a linear combination of the others.